

A Comprehensive Review of Transcranial Direct Current Stimulation (tDCS): Mechanisms, Cognitive Effects, Applications, and Future Directions

Presented By

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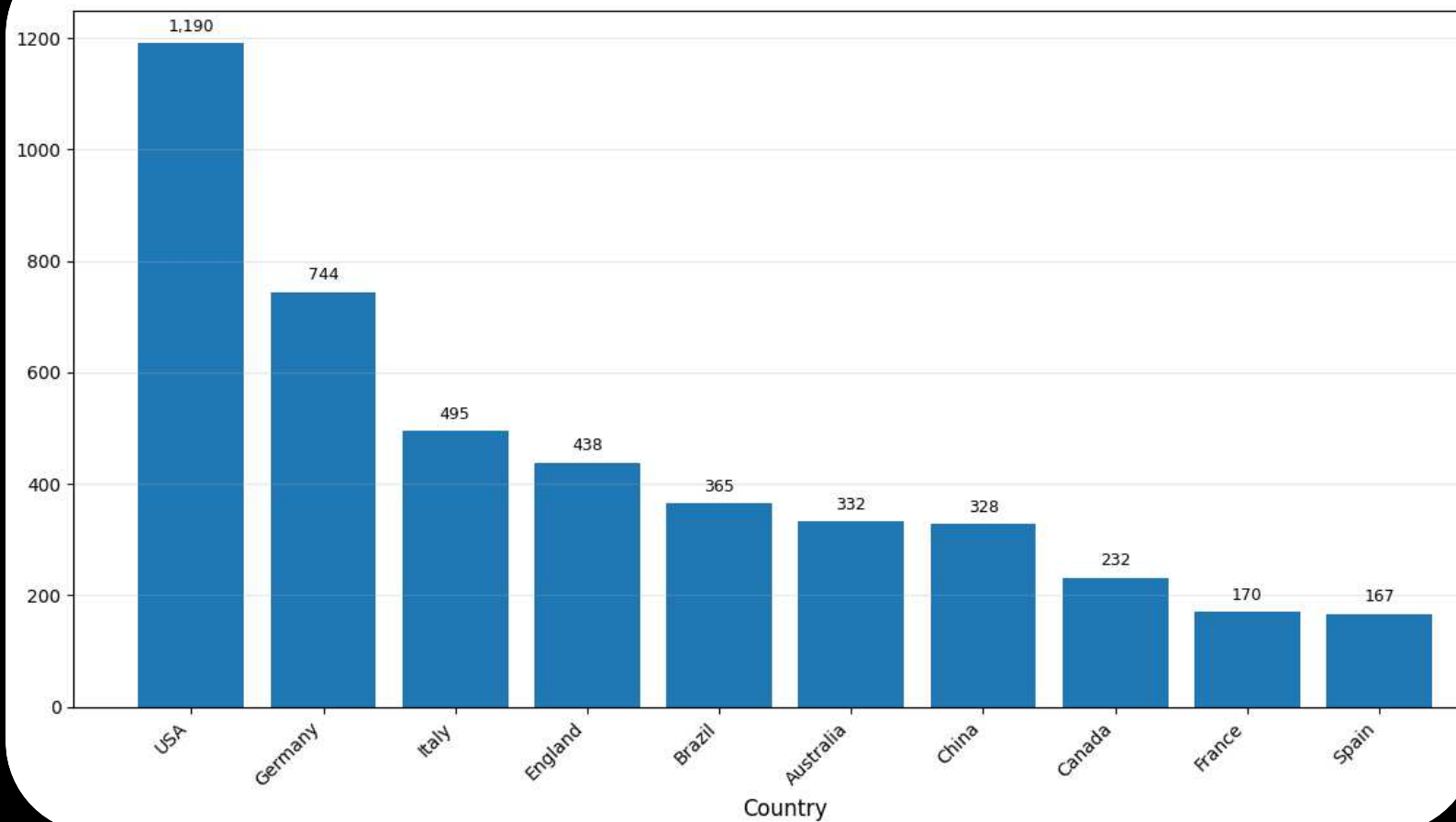
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MALAK M. N. AL-KOSHAB



Scientific Publications by Country (2000-2022)



Review Objectives:

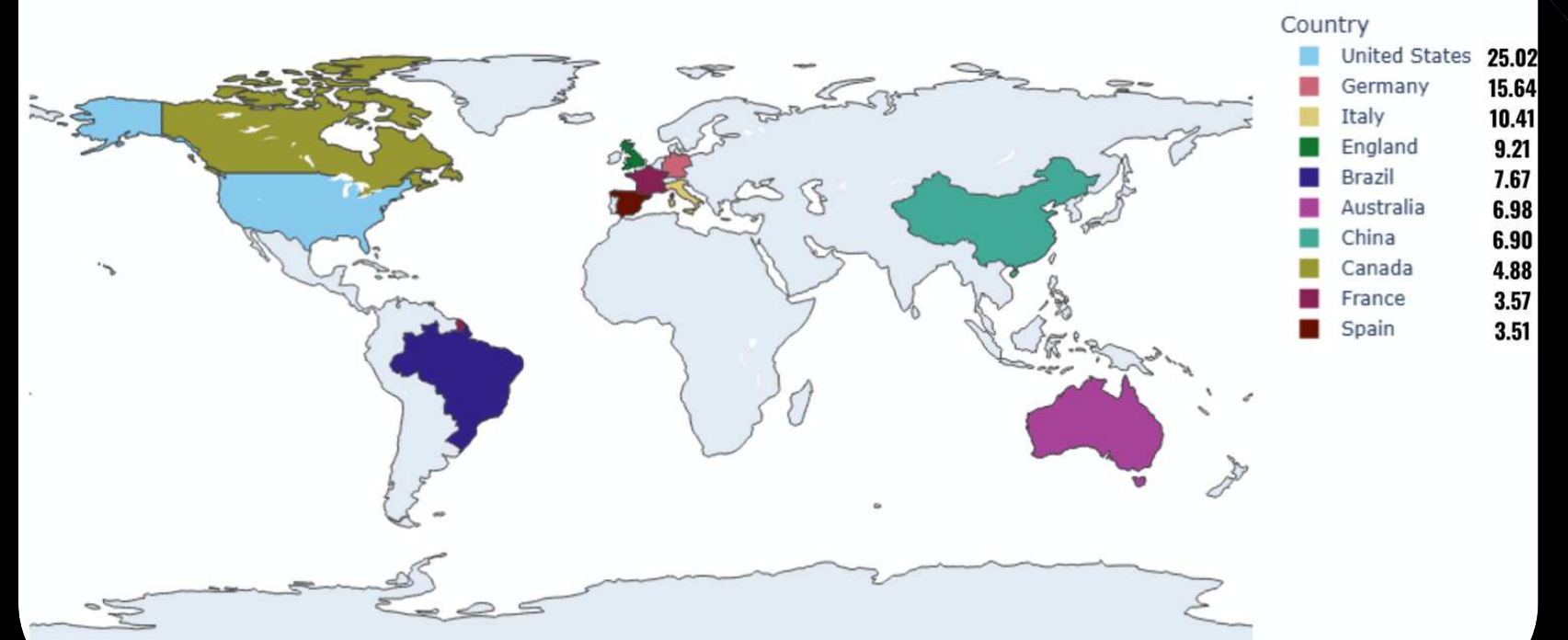
- Analyze tDCS mechanisms from cellular to neural network levels .
- Evaluate clinical efficacy across cognitive and neurological domains .
- Identify future research directions .

Introduction

Research Problem:

- Need for non-invasive brain stimulation techniques in neuroscience research and therapy .
- Challenges in standardizing stimulation protocols .

Top 10 Countries in tDCS Research by Percentage



Definition Of tDCS

A non-invasive brain stimulation technique .
Enhances cognition and supports neurorehabilitation .
Acts from cellular to network levels .
Modulates cortex via polarity .
Known for simplicity, tolerability, and neuromodulation .

IMPORTANCE

Safely modulates brain activity in awake humans .
Simple setup for research and clinical use .
Adaptable and suitable for repeated use .



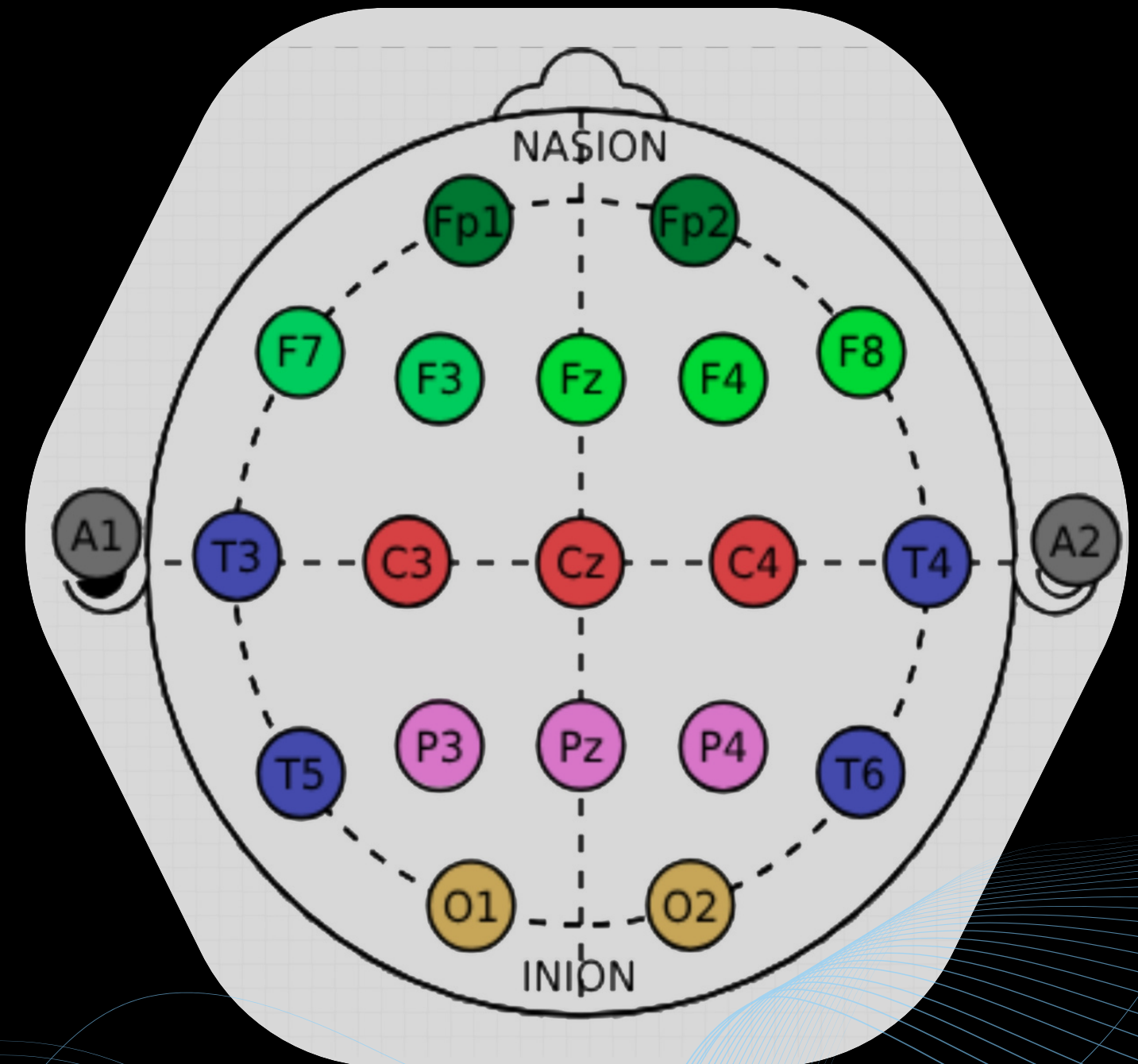
Technical Fundamentals

Stimulation Electrodes:

- 10-20 international EEG system for positioning .
- Comparison between conventional tDCS and HD-tDCS .

Stimulation parameters :

- Current intensity (1–2 mA) .
- Session duration and number (20 minutes, 15–20 sessions) .
- Neuroanatomical and genetic individual variability .



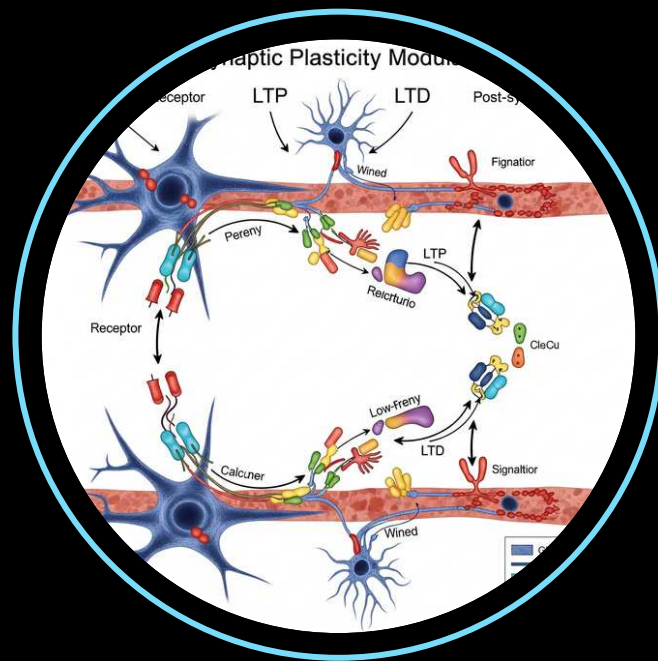
Neural Mechanisms

Polarity Effects:

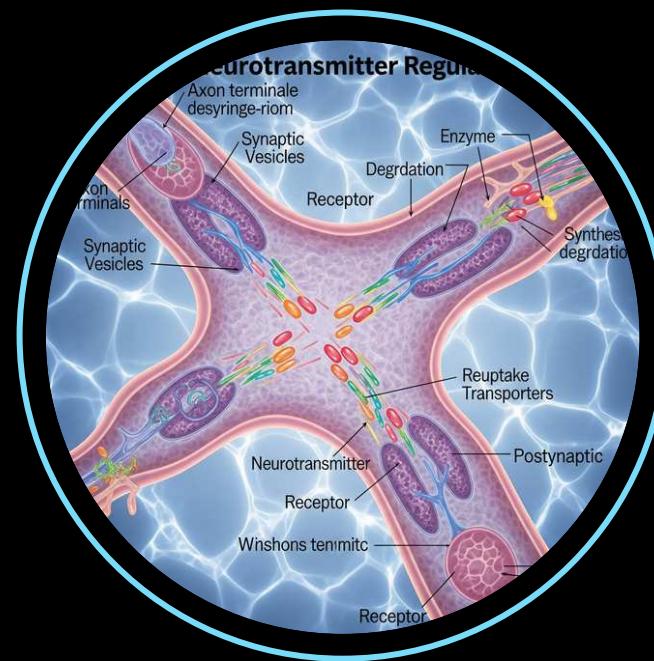
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Anode: ↑ Neural excitability via depolarization
Cathode: ↓ Neural activity via hyperpolarization

Molecular Effects:

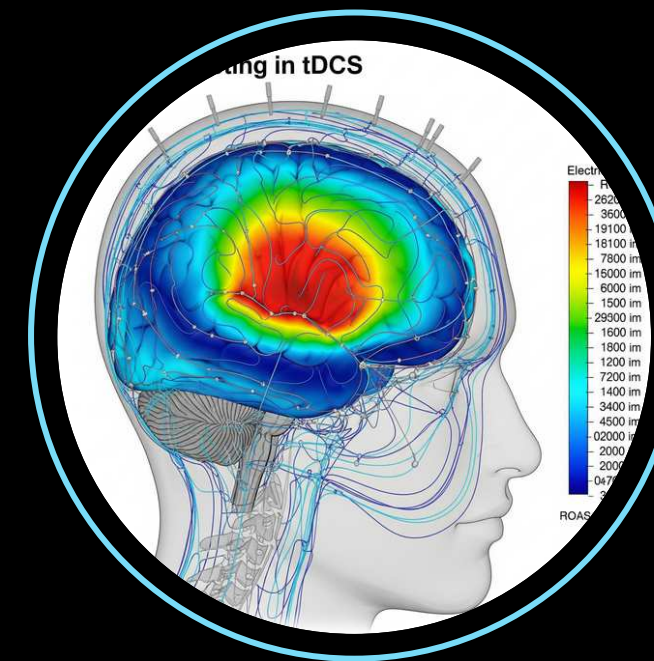


Synaptic plasticity modulation (LTP/LTD) .

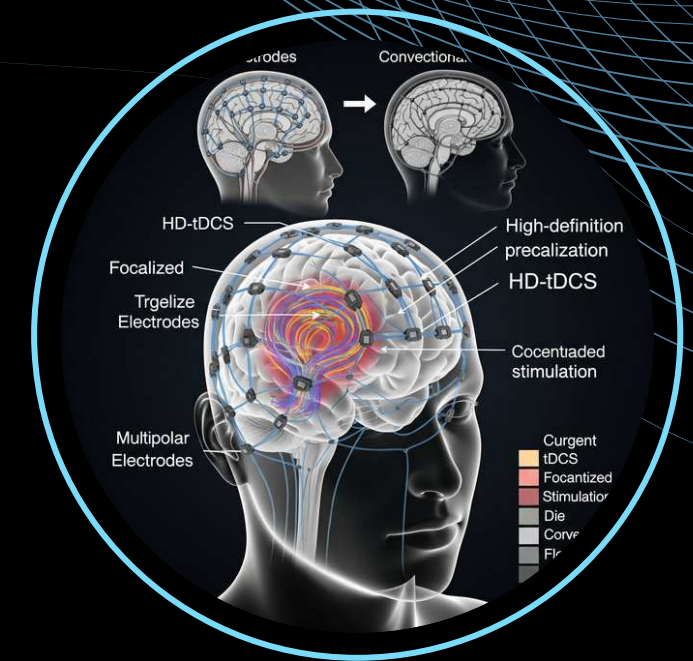


Neurotransmitter regulation:

- ↑ Glutamate (anode) .
- ↓ GABA (cathode) .



Achieving precise : targeting using ROAST .simulation



Employing multipolar: . electrode arrangements to focalize current delivery.



Safety and Ethics

Safety Profile:

Ethical Considerations:

1

First

- Common side effects
(transient):

- Skin redness (68% of studies) .
- Mild headache (35%) .
- Itching (20%) .
- Pain rating 2.21 (FACES) .

2

Second

- core ethical principles :

- 1- informed consent .
- 2- methodological transparency .
- 3- risk minimization .

3

Third

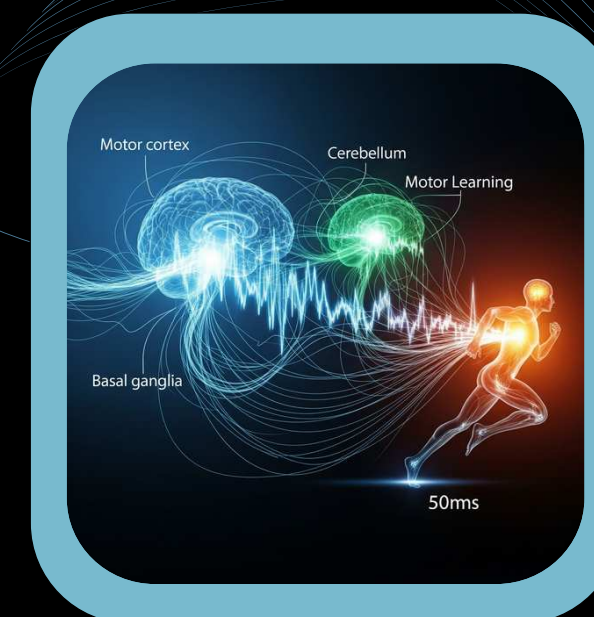
- safety protocols :

- current intensities and session durations .
- Electrode configurations and HD-tDCS arrays .

Applications



1. Cognitive Enhancement:



2. Motor rehabilitation:



3. Neurological Disorders:

Study	Target Region	Parameters	Electrode Placement	Key Findings	Adverse Effects
[13]	L/R DLPFC	2 mA, 20 min	Anode: F3, Cathode: SO	Verbal WM: ↑ naming accuracy; Visuospatial WM: ↑ performance; Retention: 2 weeks	Mild headaches; Skin flushing
[24]	Bilateral DLPFC/PPC	2 mA, 20 min	Anode: L DLPFC, Cathode: R DLPFC	Convergent thinking: ↑ CRA performance; Problem-solving: ↑ insight solutions; Analytical approaches: ↓	No adverse effects
[19]	L IFG	2 mA, 20 min	Anode: L IFG, Cathode: R Temporal	PPA variants: ↑ written naming; Generalization to untreated words; Retention: 2 months	Low pain (FACES: 2.21)
[14]	R DLPFC/PPC	1.5 mA, 20 min	HD-tDCS array	Executive vigilance: ↑ ($p < 0.05$); Parietal alpha: ↓ power; Arousal vigilance: No effect	No group differences
[25]	R TPJ/L DLPFC	1 mA, 20 min	Anode: R TPJ, Cathode: L DLPFC	Creativity: ↑ novelty/appropriateness; L DLPFC: ↑ novelty only	None reported
[21]	L DLPFC	2 mA, 20 min	Anode: F3	WM: Near-transfer gains (n-back); 9 sessions over 3 weeks	IRR: 0.8
[20]	L DLPFC	1.5 mA, 15 min	Anode: F7	WM/learning: ↑ potential; Immediate memory: No effect	Cathodal inhibition
[22]	R DLPFC	1.5 mA, 20 min	Anode: F4	Convergent thinking: ↑ usefulness; Divergent thinking: No effect	NS differences
[23]	Bilateral IFG	1.5 mA, 15 min	Anode: L IFG, Cathode: R IFG	Semantic processing: ↑ bundled trials; High-distance semantics: ↑	EEG exclusions

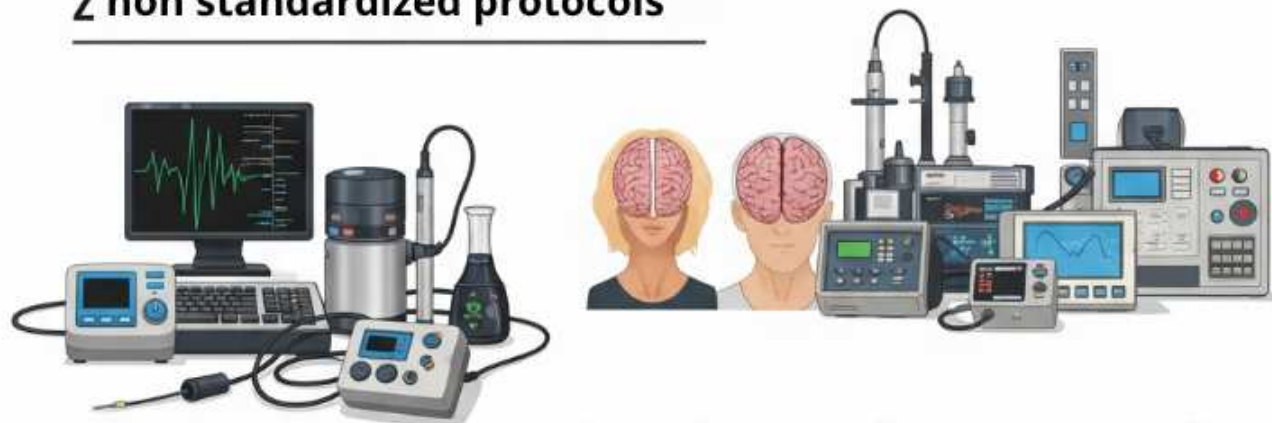
Major Challenges

Challenges facing

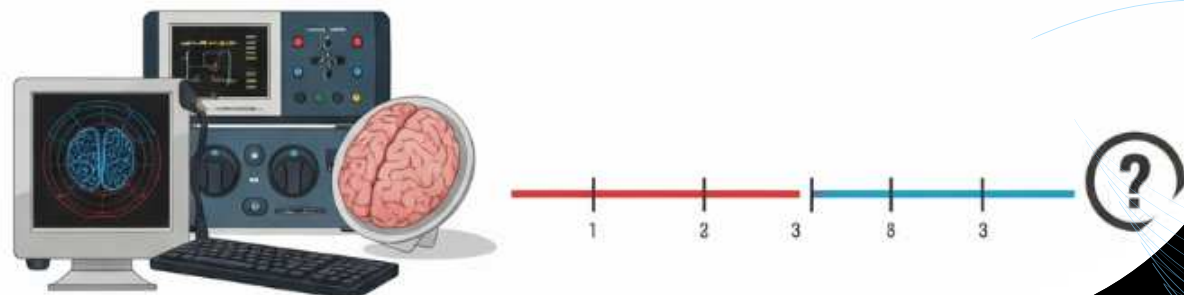
Individual Direct Current Stimulation



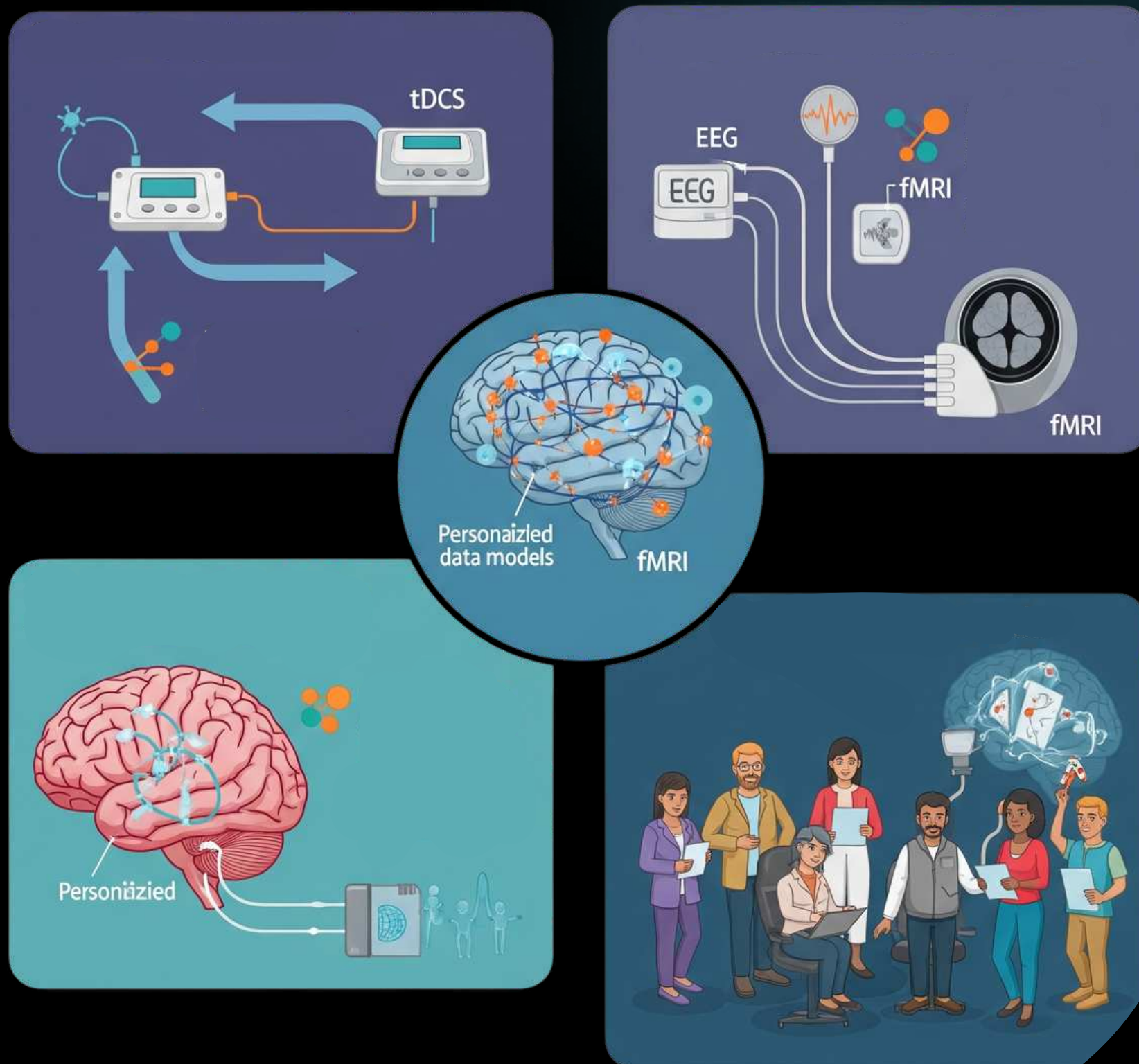
2 non standardized protocols



3. Limited scope of Long-term studies



- Individual response variability ($>300\%$) [17] .
- Lack of standardized protocols .
- Limited long-term studies .



Future Directions:

1. Closed-loop systems .
2. Integration with neurotechnologies (EEG/fMRI) .
3. Computational modeling-based personalization .
4. Large-scale clinical trials .

Conclusion

Key Findings:

1. tDCS is a promising tool with excellent safety profile .
2. Proven efficacy in cognitive enhancement and rehabilitation .
3. HD-tDCS offers improved spatial precision .

Key Findings of tDCS Research

Non-invasive
• Morin



Excellent
Safety Profile

Targeted
• Therapeutic



Cognitive
Enhancement &
Rehabilitation

Focused
• Therapeutic



Improved
Spatial Precision
of HD-tDCS

Conclusion

Future Requirements:

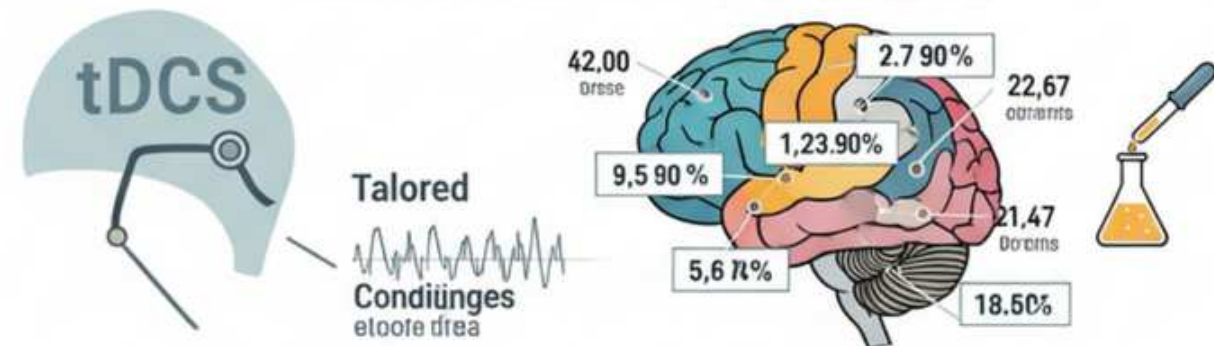
- Protocol standardization
- Personalized dosing approaches .
- Long-term longitudinal studies .

FUTURE REQUIREMENTS IN tDCS

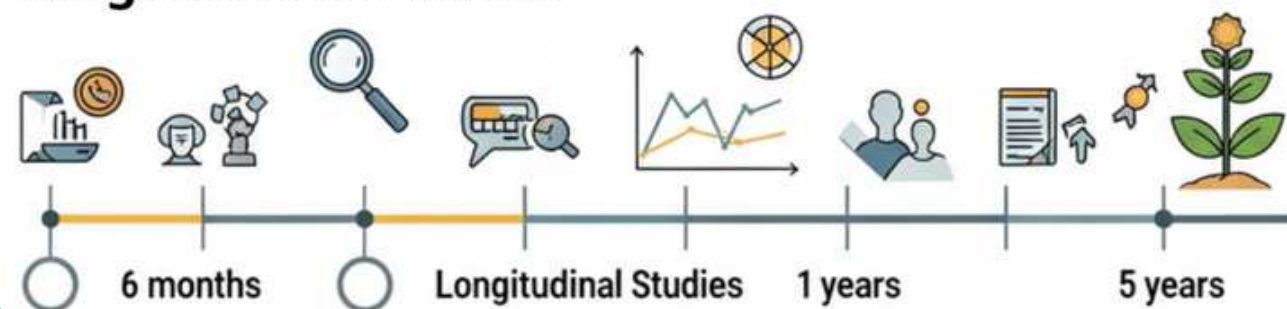
PROTOCOL STANDARDIZATION



PERSONALIZED DOSING APPROACHES



Longitudinal studies .



Thank You

